

Temperature-Dependent Optical Band Gap in CsPbBr₃, MAPbBr₃, and FAPbBr₃ Single Crystals

Giovanni Mannino, Ioannis Deretzis, Emanuele Smecca, Antonino La Magna, Alessandra Alberti,* Davide Ceratti,* and David Cahen

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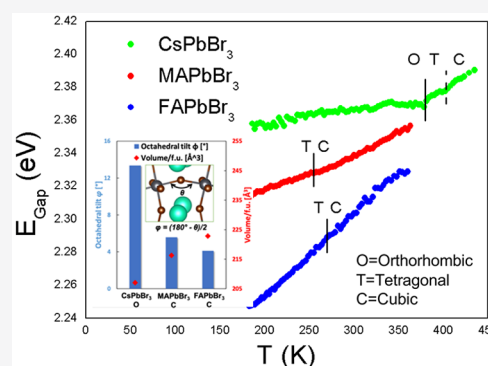
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ABSTRACT: Single crystals represent a benchmark for understanding the bulk properties of halide perovskites. We have indeed studied the dielectric function of lead bromide perovskite single crystals (MAPbBr₃, CsPbBr₃ and for the first time FAPbBr₃) by spectroscopic ellipsometry in the range of 1–5 eV while varying the temperature from 183 to 440 K. An extremely low absorption coefficient in the sub-band gap region was found, indicating the high optical quality of all three crystals. We extracted the band gap values through critical point analysis showing that Tauc-based values are systematically underestimated. The two structural phase transitions, i.e., orthorhombic–tetragonal and tetragonal–cubic, show distinct optical behaviors, with the former having a discontinuous character. The cross-correlation of optical data with DFT calculations evidences the role of octahedral tilting in tailoring the value of the band gap at a given temperature, whereas differences in the thermal expansion affect the slope of the band gap trend as a function of temperature.



The strategy for choosing the lattice composition of photoactive halide perovskites is not unidirectional. It is sometimes based on the need of avoiding organic moieties in the composition, as this is thought to be a weak point for the decomposition of the materials through the formation of volatile species.^{1–5} However, solar cells with the best performance still contain organic cations.^{6,7} It is at other times pushed by the need to increase the robustness of the inorganic lattice frame, with bromides preferred to iodides. Adding bromides increases the band gap compared to corresponding pure iodides, and this favors tandem coupling with silicon in combined photovoltaic technologies.^{8,9} Their relatively wide band gap is likewise exploited for other optoelectronic applications (LEDs and lasers). In addition to hybrid perovskites, recent developments push toward all-inorganic or Cs-containing materials.^{10–12}

Single-crystal perovskites therefore represent benchmarks for the study of the intrinsic behavior of the bulk of those materials, free from morphological effects that are instead encountered in the polycrystalline counterparts.¹³

It was also disclosed that FA-based lead iodide and bromide single crystals exhibit diffusion lengths that are close or even above 10 μm (respectively).¹⁴ Better carrier lifetime and lower dark carrier concentrations were also measured in FA-based lead iodides and bromides than in MAPbBr₃ single crystals. Furthermore, unlike the corresponding iodides, FA-based lead bromides take a stable cubic perovskite structure with space group *Pm*3̄*m* at room temperature.

Although large efforts were made to determine the UV–visible absorbance by spectrophotometric analyses, to the best of our knowledge very little is known about the optical and dielectric properties of single-crystal lead bromide perovskites.^{14–17} The literature on the optical constants of lead bromide perovskites is limited to CsPbBr₃ or MAPbBr₃ single crystals.^{18–21} In other cases, the study relies on nanocrystals rather than on millimeter-large single crystals.²²

Here we report a systematic study of the absorption coefficient and refractive index of bulk lead bromide perovskite single crystals (XPbBr₃ with X = MA⁺, FA⁺, and Cs⁺) through spectroscopic ellipsometry measurements (see [Experimental Methods](#) and the [Supporting Information](#) for further details). We show that their temperature-dependent optical properties are shaped by the cation type, which indirectly impacts on structural features like the octahedral tilting or the thermal expansion that are responsible for their optical response.

We first extract the real (ϵ_1) and imaginary (ϵ_2) part of the dielectric constant for all three perovskites ([Figure 1a](#)) from modeling the spectroscopic ellipsometry data ([Figure SI-1](#)). We identify some differences among the spectra, both in the

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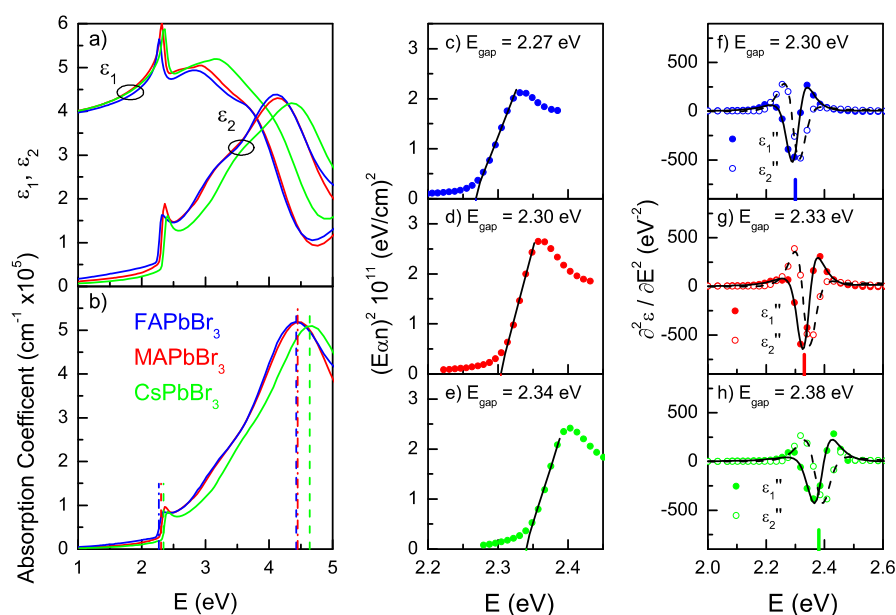


Figure 1. (a) Real and imaginary parts of the dielectric constant of the three perovskites and (b) their absorption coefficient, as derived from the data in Figure SI-1. Estimates of the gap are shown in panels c–h, using (c–e) the Tauc plot and (f–h) the critical point analysis for (c and f) MAPbBr₃, (d and g) FAPbBr₃, and (e and h) CsPbBr₃. Values of the gap are given as labels within the panels.

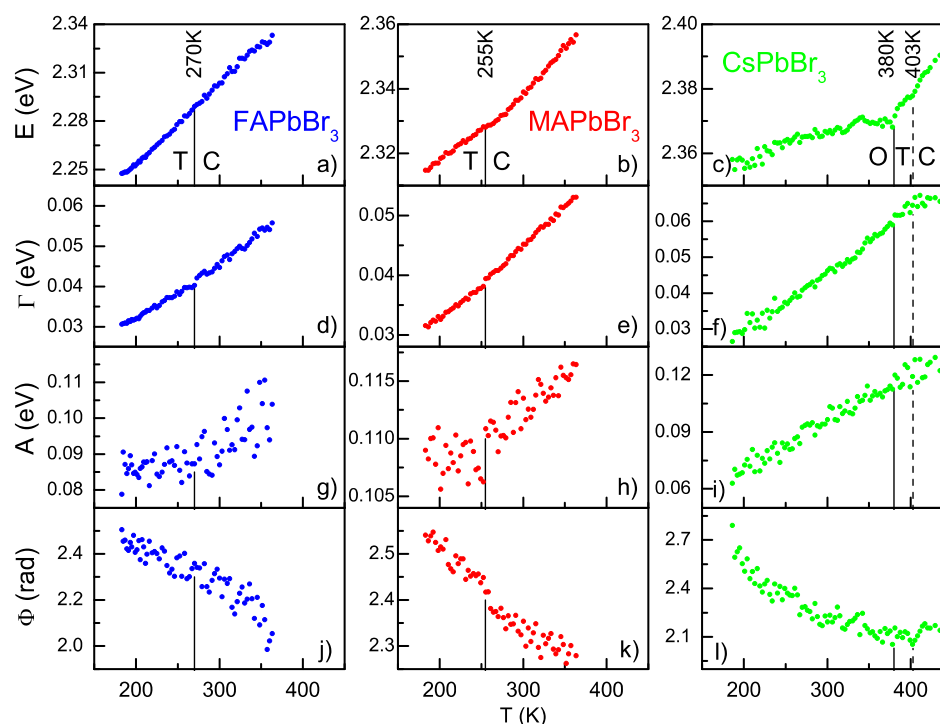


Figure 2. Critical point energy (E), broadening (Γ), amplitude (A), and phase (Φ) as extracted from fitting of eq 2 for the three XPbBr₃ (FA in blue, MA in red, and Cs in green). Continuous vertical lines represent phase-change temperatures observed from data reported here whereas dashed vertical lines are from ref 20. In the figures, the labels are T for tetragonal, C for cubic, and O for orthorhombic phase.

value of the band gap (around 2.30 eV) as well as for the main absorption peak (see ϵ_2 curve) at ~ 4 eV. The absorption coefficient is calculated according to the equation

$$\alpha = \frac{2\omega}{c} \sqrt{\frac{\sqrt{(\epsilon_1^2 + \epsilon_2^2)} - \epsilon_1}{2}} \quad (1)$$

where ϵ_1 and ϵ_2 are obtained from the optical model and ω is the light frequency. The values of the absorption coefficients

for MAPbBr₃ and FAPbBr₃ are very similar, whereas for CsPbBr₃ the main peak is blue-shifted by ~ 0.2 eV (Figure 1b). The fine differences in the gap value (E_g) may have multiple origins (e.g., the size of the X-site cations, the tilting of the inorganic octahedral network, the structural phase of each material at a given temperature, the thermal expansion coefficient) and will be hereafter discussed in detail.

Another important feature to highlight is the region where the absorption coefficient is extremely low. The sudden drop

below (at energies less than) E_{gap} occurs where the material is transparent and light travels for millimeters, thus indicating the high optical quality of the prepared materials. The finding is consistent with the long diffusion length of carriers, as found for FAPbBr_3 crystals.¹⁴

A precise evaluation of the band gap value is a major anchor for theoretical models and for comparing experimental data to judge their relevance for photonic and photovoltaic devices. A simple way to determine the gap consists of using the absorption coefficient from eq 1 and the refractive index in Tauc plots ($E\alpha(E)n(E)^2$ versus E (see Figure 1c–e).

A more accurate determination of the gap value is obtained by using the critical point (CP) analysis.^{23–26} This analysis relies on the densities of states of electron and phonon bands possessing singularities corresponding to critical points in the dielectric function (see the Supporting Information for further information).^{27–30} These points are better identified by calculating the second derivative of the measured dielectric function and finding the simultaneous best fit to the real and imaginary parts of the dielectric function, using a least-squares procedure, with the following equation:

$$\frac{\partial^2 \epsilon}{\partial \omega^2} = n(n-1)Ae^{i\Phi}(\omega - E + i\Gamma)^{(n-2)} \quad (2)$$

where E , Γ , A , Φ , and n describe the shape (energy threshold, amplitude, broadening, excitonic phase angle, and singularity dimension) of the dielectric function around the band gap energy point (see Figure 1f–h). If the gap is a discrete excitonic transition, $n = -1$.^{31–33} The fitting results of E_{gap} are shown in Figure 1f–h. By comparing these values with the ones obtained by the Tauc plot, we conclude that E_g is systematically underestimated by ~ 0.03 eV in the latter.

We then explored the optical behavior of the materials as a function of temperature to measure the band gap values over a wide range from 183 to 363 K, wherein it is expected that the XPbBr_3 samples undergo lattice changes at least from tetragonal to cubic (~ 235 K for MAPbBr_3 and ~ 265 K for FAPbBr_3).³⁴ In the case of CsPbBr_3 , an additional, orthorhombic–tetragonal transition is expected at ~ 361 K.³⁵ We accordingly extended, in this case, the temperature window up to 440 K. In all explored cases, we used a pure dry N_2 environment to prevent sample degradation due to humid air.^{5,36} The data have been rationalized using the CP analysis for better estimating not only of the E_{gap} position but also other parameters that provide important complementary information on the material evolution with temperature. The results for all fitting parameters are shown in Figure 2. A comparison of the E_{gap} position extracted from CP analysis and Tauc plots as a function of the temperature is also given in Figures SI-2 and SI-3.

Each material's data has specific features that will be discussed separately.

In the case of MAPbBr_3 , the tetragonal–cubic transition signed by the band gap is continuous. Notwithstanding this, a well-marked discontinuity is observed in the broadening A and excitonic phase Φ , whereas a drastic change of slope (from decreasing to increasing) is observed in the amplitude parameter. These significant modifications are also visible in the change of slope of the E_{gap} at ~ 255 K.

In the case of FAPbBr_3 , the A and Φ discontinuity is less evident but still present as corroborated by a discontinuity in the broadening parameter. Moreover, the amplitude, constant for lower temperatures, starts increasing at the same temper-

ature of the discontinuity in the broadening parameter, at ~ 270 K. At the same time, the E_{gap} increases linearly over the whole temperature range, and no discontinuity is observed at the transition temperature.

As a general paradigm, the FA- and MAPbBr_3 show a unique and coherent picture of phase transformation from tetragonal to cubic by increasing the temperature. The parameter continuity is a typical behavior observed in order–disorder phase transitions in halide perovskites.^{37,38}

In the case of CsPbBr_3 , the crystal is in the orthorhombic phase in the low-temperature range. Its E_{gap} undergoes a jump and a sudden change of slope when the lattice is transformed to tetragonal at ~ 380 K. In contrast to the slow (FA- and MAPbBr_3) and well-marked (MAPbBr_3) transitions from tetragonal to cubic phases, not much can be said about the same transition in CsPbBr_3 . The upper bound of the temperature window for the tetragonal phase is only 20 K before the lattice becomes cubic, and such an interval is too narrow to extract accurate fitting parameters considering that intermediate phases might exist.⁵⁴ However, for the sake of completeness, we added the expected transition point from literature data at 403 K.²⁰

Besides the highlighted differences, a common feature to all three materials is that the excitonic phase (Φ) decreases with temperature, empirically indicating a decrease of excitonic interactions.^{27–30} Finally, we observe that, compared to literature data, the phase-change temperatures provided by spectroscopic ellipsometry are shifted upwards by 20 K for MA- and Cs- and by 5 K for FAPbBr_3 .

Although the trend of the E_{gap} versus T and its value near 0 K are often fitted with the Bose–Einstein equation to account for electron–phonon interactions,^{27–30} MAPbBr_3 and FAPbBr_3 are still in the tetragonal phase within the investigated T -range while in CsPbBr_3 the transition to orthorhombic is not linear (see also Figure SI-3).^{39,40} A linear fit in this region would overestimate $E_{\text{gap}}(0)$ and would contradict what was reported in ref 22 (see Figure 3). To properly apply the Bose–Einstein equation, the sample should be cooled down near 0 K.^{21,22,41} Under these circumstances, the relative weight of thermal expansion and electron–phonon coupling could be evaluated, as in the case of MAPbI_3 .⁴²

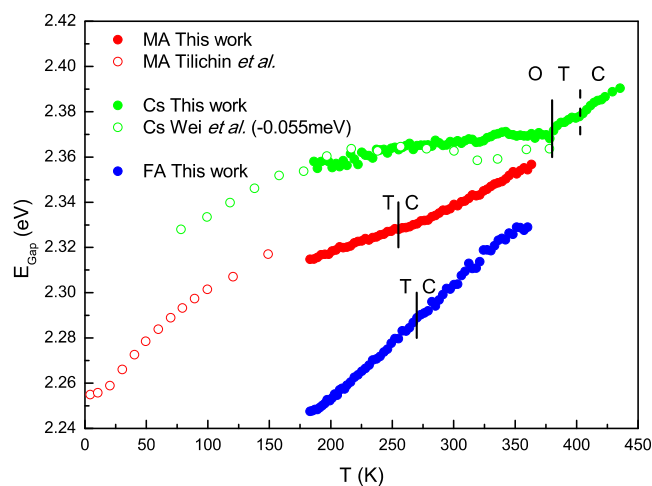


Figure 3. Temperature dependence of E_{gap} in comparison with data from Wei et al.²² and Tilchin et al.²¹ The labels are T for tetragonal, C for cubic, and O for orthorhombic phase as in Figure 2.

The temperature dependence of the band gap was fitted using $E_g(T) = A + B \cdot T$. All the fitting values are summarized in Table 1. FA- and CsPbBr₃ show a linear behavior over the whole

Table 1. Values of Parameters Used to Fit Data in Figure 3a–c^a

T–C phases	A (eV)	B ($\times 10^{-4}$ eV K ⁻¹)
FAPbBr ₃	2.154 $\pm$ 0.001	4.98 $\pm$ 0.04
MAPbBr ₃ (183–255 K)	2.279 $\pm$ 0.001	1.91 $\pm$ 0.04
MAPbBr ₃ (255–363 K)	2.256 $\pm$ 0.001	2.75 $\pm$ 0.03
CsPbBr ₃ (380–435 K)	2.242 $\pm$ 0.001	3.41 $\pm$ 0.1

^aValues refer to the interval wherein each XPbBr₃ is in the tetragonal (T) and cubic (C) phases.

range of the tetragonal and cubic phases. The E_g dependence of MAPbBr₃ deviates from linearity at the transition temperature from tetragonal to cubic, making necessary the use of two distinct linear equations.

Finally, we compare our data to the very few other data available in the literature dealing with the specific task of evaluating the E_{gap} .^{21,22} For MAPbBr₃, a phase transition from orthorhombic to tetragonal is expected at ~ 150 K where, similar to CsPbBr₃, some discontinuity is expected. Therefore, data reported here and those in ref 21 are in excellent agreement. In the case of CsPbBr₃, data collected in ref 22 obtained on quantum dots have been shifted to match our data in order to account for the effect of quantum confinement.

As discussed earlier, the origins of the fine differences in the band gap, optical constants, and temperature behavior of the three XPbBr₃ can be attributed to structural features related to the X-site cations. There are three characteristics of the band gap behavior of Figure 3 that are of interest.

The first one regards the value of the band gap, which appears to be higher for the CsPbBr₃ perovskite than for MAPbBr₃, which is higher than for FAPbBr₃.

The second issue regards the slope of the band gap's temperature dependence. Here, when considering equivalent phases for the three materials (either tetragonal or cubic), FAPbBr₃ shows the steepest band gap increase with increasing temperature, followed by CsPbBr₃ and then MAPbBr₃ (cubic phase).

The third issue that deserves discussion is the quasi-continuous behavior of the band gap trend at the tetragonal–cubic phase transition, in contrast with the discontinuous behavior of the orthorhombic–tetragonal transition (seen here for the CsPbBr₃ perovskite within the studied temperature range).

We now discuss these aspects in detail. It is already known from lead iodide perovskites that two main factors affect the value of the band gap: the change in the unit cell volume (through a variation of the lattice constants) and the octahedral tilting of the inorganic framework.^{43,44} The first can be evaluated following ref 42 and using the thermal parameters (the volumetric expansion coefficient (α_v) and the bulk modulus (β) where available) reported in refs 40 and 45–51.

As we can see from the calculated data (compare $\left[\frac{\partial E_g}{\partial T}\right]_{\text{TE}}$ in Table 2 with B in Table 1), the variation of the band gap due to the temperature-related volume increase is not sufficient to explain our findings. We indeed evaluate the contribution of the octahedral tilting with the aid of the density functional

Table 2. Literature Values Used to Estimate the Variation of the Band Gap Due to the Thermal Expansion^a

	α_v ($\times 10^6$ K ⁻¹)	β (GPa)	$\frac{\partial E_g}{\partial P}$ (eV/GPa)	$\left[\frac{\partial E_g}{\partial T}\right]_{\text{TE}}$ ($\times 10^{-4}$ eV K ⁻¹)
FAPbBr ₃	150	12.3	−0.20	3.69
MAPbBr ₃	33	19.6	not available	—
CsPbBr ₃	50	15.8	−0.28	2.21

^aThe relationship used is $\left[\frac{\partial E_g}{\partial T}\right]_{\text{TE}} = -\alpha_v \beta \frac{\partial E_g}{\partial P}$. In the case of

MAPbBr₃ the value of $\frac{\partial E_g}{\partial P}$ is not reported. MAPbBr₃ has a phase transition at 0.1 GPa which do not allow obtaining accurate data.

theory (DFT, see the Supporting Information). Figure 4a shows the band structures of CsPbBr₃ in the cubic and the orthorhombic phase at 0 K, where we observe an increase in the band gap value for the orthorhombic symmetry.

We note that the main geometrical feature distinguishing the two phases is the tilt angle, φ , for the $[\text{PbBr}_6^{4-}]$ octahedra, which is higher for the orthorhombic phase. Figure SI-4 shows the impact of either octahedral tilting (cubic $\rightarrow$ orthorhombic) or isotropic volume variation (of the cubic phase) on the band edge energies of the CsPbBr₃ perovskite, relative to the vacuum level. An isotropic expansion of the unit cell volume increases both the ionization potential and electron affinity, i.e., shifts the valence band maximum (VBM) and conduction band minimum (CBM) toward more negative energies relative to the vacuum level. However, because the VBM changes more than the CBM, the band gap increases. Similarly, an isotropic compression of the unit cell reduces the band gap value. Octahedral tilting increases the band gap by shifting the VBM away from the vacuum level, toward more negative energies, and the CBM toward more positive energies. We expect that these are general trends for both hybrid and inorganic XPbBr₃, as the band gap in all cases is defined by antibonding Pb and halogen orbitals, whereas states that derive from the X-cations are well-removed in energy from the band gap region.⁵² It is therefore important to keep in mind that the band gap of a XPbBr₃ sample can increase by either an isotropic volume expansion or by the tilting of the $[\text{PbBr}_6^{4-}]$ octahedra.

We now compare the structural parameters of all three perovskites in their corresponding room-temperature phases (i.e., orthorhombic for CsPbBr₃ and cubic for MAPbBr₃ and FAPbBr₃) by means of the DFT calculations (Figure 4b). We find the result of our calculations to agree with the experimental data that CsPbBr₃ has the smallest volume per formula unit, followed by MAPbBr₃ and then FAPbBr₃, reflecting the X-cation sizes.^{40,53–56} On the basis of the volumetric data, it is not possible to explain the differences of the experimental band gap values for the three perovskites (Figure 3). We then observe the tilting angle φ (Figure 4b). Here, the highest value is obtained for CsPbBr₃, followed by MAPbBr₃ and FAPbBr₃. We note that a local tilting of the $[\text{PbBr}_6^{4-}]$ octahedra can take place also in the cubic phase. Macroscopically though its effect is canceled out because of the random orientations or positions of the X-cations within the inorganic cage. Considering these results, it appears that octahedral tilting has a primary role in shaping the band gap value for the three perovskites and explains the differences observed in Figure 3.

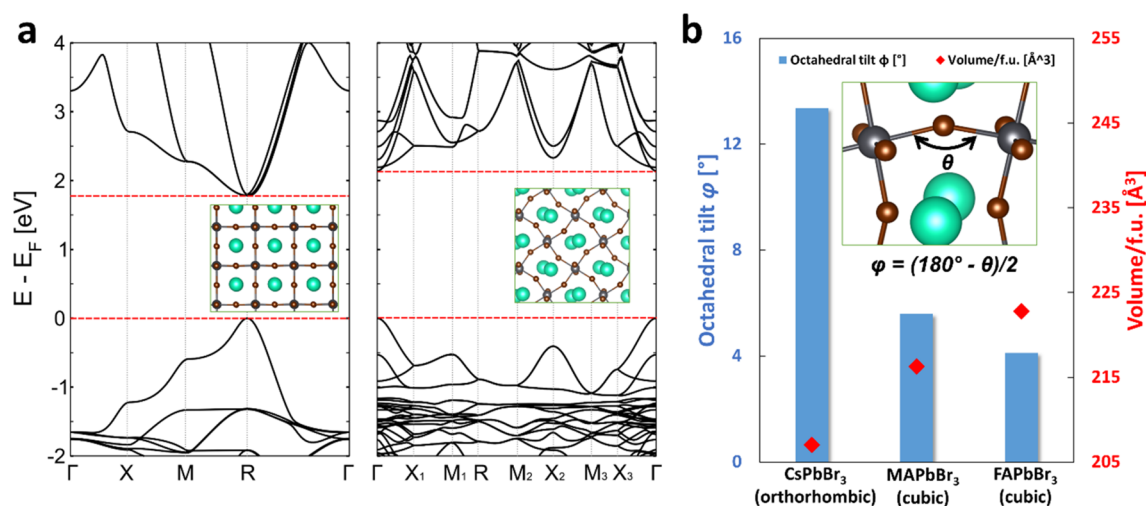


Figure 4. (a) Differences in the calculated band gap of CsPbBr₃ in the cubic (left) and orthorhombic (right) phase for models representing the structures at $T = 0$ K. (b) Calculated octahedral tilting angle ϕ (columns) and volume per formula unit (points) for orthorhombic CsPbBr₃, cubic MAPbBr₃, and cubic FAPbBr₃ (corresponding to the phases of each material at room temperature).

The slope of the band gap value as a function of the temperature cannot be deduced by our DFT calculations, as these reflect a “frozen” static picture. It is most likely related to the thermal expansion of the three perovskites (considering the volumetric data reported in refs 53, 54, and 40), if we consider separately each perovskite phase. Here, it is important to note that the optically continuous tetragonal–cubic transition (at least for the band gap value, see Figure 2) indicates also a structural relationship between the two phases. In contrast, the discontinuity in the orthorhombic–tetragonal transition denotes a sharp structural change related to the X-cations, which influences the vibrational properties of the system.⁵⁷

In conclusion, we studied the phase transition of three XPbBr₃ single crystals, namely, MAPbBr₃, CsPbBr₃, and (for the first time) FAPbBr₃ in the range of temperature 183–440 K by spectroscopic ellipsometry under nitrogen atmosphere. Optical data were extracted by applying the critical point analysis, with special regard to the band gap region. The extremely low value of the absorption coefficient in the subgap region certifies a small scattering inside the crystal and indeed the high optical quality of the materials. We propose a close relationship connecting the structural parameters with the optoelectronic properties of the three kinds of crystals. The combined experimental–theoretical analysis indicates two important parameters for the band gap determination in XPbBr₃: octahedral tilting, which is fundamental for the band gap value, and the thermal expansion, which likely impacts the slope of the band gap as a function of the temperature.

EXPERIMENTAL METHODS

Sample Preparation. MAPbBr₃, FAPbBr₃, and CsPbBr₃ single crystals were grown by the method reported in the supplementary information of ref 13. We add here details of the synthesis that help to obtain with the antisolvent method crystals larger than those created before. To that end, we cover the container of the perovskite precursors with filtering paper. This avoids unwanted dropping of antisolvent that condensed on the crystallization chamber walls. Any antisolvent drop falling in the precursor solution induces the formation of multiple seeds, which reduces average crystal size. The crystals

were conserved in boxes containing nitrogen with silica gel. See the Supporting Information for details.

Spectroscopic Ellipsometry. We used a V-VASE, J. A. Woollam spectroscopic ellipsometry instrument for optical characterization. Measurements were collected at four angles, 60°–65°–70°–75°, below and above the Brewster angle, over a wide range of wavelengths 245–1240 nm (1–5 eV). We explored also a wide temperature range (180–440 K) varying the temperature with an Instec MK100 heater/cooler system with an accuracy of 0.1 °C. See the Supporting Information for further setup details and data modeling.

X-ray Diffraction. High-resolution X-ray diffraction analyses have been performed on single-crystal perovskite samples using a D8Discover Bruker AXS diffractometer by focusing the beam over an area of 2 mm in diameter and using a Ge022 monochromator to remove the k_2 component. Data are shown in the Supporting Information (Figure SI-5).

Density Functional Theory. We performed density functional theory calculations as implemented in the plane-wave Quantum Espresso code.⁵⁸ We used the Perdew–Burke–Ernzerhof (PBE) implementation of the generalized gradient approximation for exchange and correlation along with scalar relativistic ultrasoft pseudopotentials.⁵⁹ See the Supporting Information for details.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpclett.0c00295>.

Method details (section 1); ellipsometric data treatment (section 2 and related figures (Figures SI-1 and SI-2); comparison between Tauc and CP gap values (Figure SI-3); theoretical evaluation of valence and conduction band variation due to compression and expansion in different bromides lattices (Figure SI-4); diffraction pattern of the FAPbBr₃ bulk crystal (Figure SI-5) (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Alessandra Alberti – CNR-IMM, 95121 Catania, Italy;

orcid.org/0000-0002-4103-6208;

Email: alessandra.alberti@imm.cnr.it

Davide Ceratti – Department of Materials and Interfaces,

Weizmann Institute of Science, Rehovot 76100, Israel;

Email: davide-raffaele.ceratti@weizmann.ac.il

Authors

Giovanni Mannino – CNR-IMM, 95121 Catania, Italy

Ioannis Deretzis – CNR-IMM, 95121 Catania, Italy;

orcid.org/0000-0001-7252-1831

Emanuele Smecca – CNR-IMM, 95121 Catania, Italy

Antonino La Magna – CNR-IMM, 95121 Catania, Italy

David Cahen – Department of Materials and Interfaces,

Weizmann Institute of Science, Rehovot 76100, Israel;

orcid.org/0000-0001-8118-5446

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.jpclett.0c00295>

Notes

The authors declare no competing financial interest.

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